

Systems Engineering for Deep Learning

Course Project

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Amnesiatic Networks: Post-hoc Machine Unlearning through Local Synaptic Plasticity

Code: https://github.com/Vishal-83-007/CS6886_Course_Project

1 Introduction

Machine unlearning asks a trained model to behave as if some of its training data had never been seen, for example after a data-deletion request. The reference answer is a *Gold* model retrained from scratch without that data; unlearning methods try to approximate it at a fraction of the cost. Almost all existing methods are gradient based. We study unlearning in a spiking neural network (SNN) trained *without* gradients, by unsupervised spike-timing-dependent plasticity (STDP), where the only information available about how each synapse was learned is local: pre- and post-synaptic activity.

We train a Diehl & Cook network [1] on MNIST and Fashion-MNIST and remove (i) two whole classes, 6 and 9 (digits 6/9; Shirt and Ankle boot), and (ii) a random 10% of the training images of those classes. We implement four post-hoc methods (A–D) using saved neuron labels or local plasticity signals, compare their available evaluations with Gold, and test B/D for durability under continued learning.

2 Prior work

Network. Diehl & Cook [1] showed that conductance-based leaky integrate-and-fire neurons with a power-law STDP rule, lateral inhibition and an adaptive threshold θ learn digit prototypes without labels; neurons are labelled afterwards by the class they respond to most. ASP [2] adds a weight leak so that rarely reinforced synapses decay, letting an SNN forget old inputs in non-stationary streams; controlled forgetting with dopaminergic modulation [3] recycles under-used neurons. Both forget *passively*; neither removes chosen data on request.

Unlearning. Retraining is the gold standard [5]. Selective Synaptic Dampening (SSD) [4] is post-hoc and retraining-free: it scores every parameter by its (diagonal Fisher) importance for the forget set D_f and the full training set D, and shrinks parameters that are much more important for D_f. Chundawat et al. [6] argue that the goal is to match Gold, not to drive forget accuracy to zero, since an over-forgetting model leaks membership (the "Streisand effect"). Recent work shows that much unlearning is reversible: fine-tuning on retained data alone restores forgotten knowledge in LLMs [7] and image classifiers [8].

Table 1: Baseline test accuracy (%) versus published STDP-SNN benchmarks and a high-performing supervised STDP approach. Architectures, training budgets and readouts differ; these are contextual comparisons.

Method	Training / architecture	MNIST	Fashion
Diehl & Cook [1]	Unsupervised; 6400 excitatory	95.00	–
Touda & Okuno [11]	Unsupervised; 400 excitatory	88.84	68.01
Fahey et al. [12]	Unsupervised SMIN; 800 neurons	92.45	78.59
Goupy et al. [13]	STDP-CSNN + supervised NCG	98.92	88.72
Ours	Unsupervised; 800 excitatory	89.37	74.28

Our entries use the saved full 10,000-image baseline evaluations. Touda–Okuno use one training epoch; we use three. Fahey et al. entries are best accuracies (not means). Goupy et al. report means (standard deviations 0.07/0.23 points) for S2-STDP+NCG with an STDP-CSNN feature extractor and five neurons per class. "–" = not reported.

Baseline suitability. Our accuracy exceeds the comparable 400-neuron unsupervised benchmark on both datasets, while using twice the excitatory neurons and more training. It is 3.08/4.31 points below the best reported 800-neuron SMIN results, and below the larger Diehl–Cook network and supervised deep STDP system. This places our baseline within the range of useful shallow STDP models for studying unlearning; all unlearning methods start from this same trained model.

Our contribution. We adapt SSD to an SNN by replacing Fisher importance with a *hypothetical-STDP* importance (Method B), turn ASP's leak into a request-gated leak with retained replay (Method D), and evaluate them together with label-guided silencing (A) and anti-STDP replay (C) against Gold, including a B/D relapse test in which the network keeps learning.

3 Methods and implementation

Network and training. $N = 800$ excitatory and 800 inhibitory neurons; Poisson input (pixel/4 Hz, 350 ms per image + 150 ms rest, up to six retries with +32 Hz if fewer than 5 spikes);

STDP $\Delta w = \eta\,(x_{\text{pre}} - x_{\text{tar}})(w_{\max} - w)^{\mu}$ at post-synaptic spikes ($\eta = 10^{-4}$, $x_{\text{tar}} = 0.4$, $\mu = 0.9$); adaptive threshold θ (+0.05 mV per spike, $\tau_\theta = 10^7$ ms); three epochs over all 60,000 images. On Fashion-MNIST θ saturated and 19% of images produced no spikes (42.8% accuracy); normalising every neuron's input weights to sum to 39 before each training image (Diehl & Cook's released code uses 78) fixed this (74.3%). We moved the whole image/retry loop into compiled C++ (Brian2 `cpp_standalone` [9]), about 10× faster than the Python loop; all runs used single CPU cores on the IITM AQUA cluster. Gold models are retrained with the forgotten images removed from both training and neuron labelling.

Evaluation. A frozen network (STDP and θ off) is evaluated with fixed neuron labels. We report (1) *deployed* accuracy with the model's fixed labels (the baseline's labels for whole-class unlearned models), split into retained and removed classes, and, where measured, (2) a *recoverability probe*: neurons relabelled using 500 labelling images per class from all ten classes, i.e. how much 6/9 information an observer with labelled examples could still read out. For B and D, hyper-parameters were chosen on a fixed tuning half of the test set (500 per class) by the mean absolute gap to Gold over these four numbers; results are reported on the other half. The supplied A/C runs instead report fixed-label accuracy on all 10,000 test images, without recoverability or relapse measurements; these entries are explicitly marked and are not used for the four-number Gold-gap ranking. C is reported descriptively at scale $s_u = 0.5$ after five passes, rather than selected by that ranking.

Method A: label-guided neuron silencing. Neurons whose saved label is 6 or 9 are prevented from firing by a per-neuron threshold override (+1 V), verified to emit no spikes; weights are unchanged.

Method B: SSD with hypothetical-STDP importance. With learning frozen, each training image x yields the update STDP *would* have made, $U_{ij}(x)$, accumulated as $I_{ij}(S) = \frac{1}{|S|}\sum_{x\in S} U_{ij}(x)^2$ per class (one 60,000-image pass, reusable for any later request). Synapses with $I_{ij}(D_f) > \alpha I_{ij}(D)$ are scaled by $\beta_{ij} = \min(\lambda I_{ij}(D)/I_{ij}(D_f), 1)$. Since $D \supset D_f$, $I(D_f)/I(D) \le |D|/|D_f| \approx 5$ for two of ten classes, so SSD's usual $\alpha = 5$–50 selects nothing; we sweep $\alpha \in \{1.5, 2, 3, 4\}$, $\lambda \in \{0, 0.1, 0.5, 1\}$.

Method C: anti-STDP and LTD-only replay. The forget images are replayed with the sign of the baseline STDP update reversed (C1) or with only the potentiating part negated (C2), for 1, 2 or 5 passes. Each event update is multiplied by s_u and clipped to $[0, 1]$; C1 can increase weights when the original rule would depress them, whereas C2 only decreases weights. Thresholds and inhibition are frozen, weight normalisation is disabled, and each image receives one 350 ms exposure plus 150 ms rest, without retries or updates during rest. Whole-class runs cover both variants and scales $0.1, 0.5, 1.0$; the subset experiment uses scale 0.5 only. Every trial starts independently from the original baseline.

Method D: request-gated leak with retained replay. R retained images are replayed with normal STDP while gated weights decay, $w \leftarrow (1 - \lambda_u)w$ after each image; θ is frozen. The gate is either every synapse of the neurons labelled 6/9 (*neuron gate*; 166 of 800 MNIST neurons) or Method B's selection (*synapse gate*). On Fashion-MNIST replay keeps the training normalisation for ungated synapses only; without this, partly gated neurons grew without bound and the network collapsed to one class.

Relapse test. Starting from each audited B/D model, we replay 20,000 retained images with STDP *and* adaptive θ on, then re-measure removed-class accuracy with the baseline's labels. This models a deployed SNN that keeps learning.

Cost. Retraining Gold took 13.7 h (MNIST) and 16.6 h (Fashion) on one core. D's neuron gate needs only the saved labels and 0.39 h / 0.13 h of replay (≤3% of Gold). B's dampening takes seconds but needs the importance pass (6.6–9.1 h, about half of Gold); it is computed once per model and stores every class, so further requests are nearly free, and it could be accumulated during training at negligible cost.

4 Evaluation

Table 2: Whole-class forgetting of 6/9: shared accuracy metrics (%). Retained/Removed = test classes other than 6/9 / classes 6/9. †A/C use the full test set; other rows use the final half.

	MNIST		Fashion-MNIST	
Model	Retained	Removed	Retained	Removed
Baseline	89.1	87.6	77.0	60.0
Gold	92.1	0	81.1	0
A: silencing†	92.0	0	81.1	0
B: SSD, $\lambda = 0.5$	91.7	0	80.9	2.1
B: SSD, $\lambda = 0.1$	91.6	0	80.2	1.8
C1: anti-STDP†	91.0	30.7	66.3	13.7
C2: LTD-only†	90.3	30.5	69.3	14.4
D: neuron gate	91.5	0	81.3	0
D: synapse gate	91.5	0	80.8	1.3

C uses $s_u = 0.5$, five passes. B uses $\alpha = 2$. D: neuron gate $\lambda_u = 0.001$, $R = 5000$ (MNIST) / $\lambda_u = 0.005$, $R = 1000$ (Fashion); synapse gate $\alpha = 2$, $\lambda_u = 0.005$, $R = 1000$. Costs are reported in Section 3; recorded timing boundaries differ.

Comparison on common metrics. Table 2 shows that A and D's neuron gate most closely reproduce Gold's deployed whole-class behaviour: both give zero removed-class accuracy while preserving retained accuracy. B also approaches Gold, with 0–2.1% removed-class accuracy. C leaves substantially more removed-class recognition and damages retained-class performance on Fashion-MNIST. These shared metrics alone do not establish a clear winner among A, B and D, particularly given the different test splits; A's unchanged weights also mean that suppression alone does not establish erasure.

Further investigation of B and D. B and D looked promising in this comparison as our novel unlearning adaptations of two separate papers: SSD [4] motivates B's selective dampening, with Fisher importance replaced by hypothetical-STDP importance, while ASP [2] motivates D's request-gated leak with retained replay. Their ability to approach Gold using local plasticity signals motivated further investigation of recoverability and durability under continued learning. Table 3 reports these additional measurements separately.

Table 3: Additional B/D audit (%). Probe: retained/removed accuracy after all-class relabelling, final test half. Relapse: removed accuracy after 20,000 adaptive retained replay images, tuning half.

	MNIST		Fashion-MNIST	
Model	Probe ret./rem.	Relapse	Probe ret./rem.	Relapse
Baseline	89.1 / 88.5	94.5*	77.2 / 59.4	62.4*
Gold	84.1 / 76.5	–	71.1 / 69.4	–
B: SSD, $\lambda = 0.5$	82.3 / 73.9	0.6	70.4 / 62.9	33.8
B: SSD, $\lambda = 0.1$	82.0 / 71.7	0.1	70.2 / 60.2	10.2
D: neuron gate	83.5 / 79.8	**0.0**	72.0 / 59.6	**0.0**
D: synapse gate	82.1 / 72.6	0.0	71.2 / 59.5	6.9

*Baseline fine-tuning on retained data alone, without an unlearning step. Gold relapse was not measured. Settings match Table 2.

Closeness to Gold. Across the deployed and probe accuracies, B/D have mean absolute Gold gaps of 1.1–1.9 points on MNIST and 2.4–3.2 on Fashion.

Recoverability. Relabelling tests whether class information can still be read out despite zero deployed removed-class predictions. Gold itself recognises 76.5%/69.4% of removed MNIST/Fashion classes; B/D recognise about 72–80%/60–63%. Retained-class neurons learn overlapping features, so even retraining does not eliminate all 6/9 recognition. The probe must therefore be interpreted relative to Gold rather than a zero target.

Relapse under continued learning. Retained-only replay raises the unmodified baseline's removed accuracy to 94.5%/62.4%. After B, dampened detectors can revive as adaptive thresholds decay: on Fashion, $\lambda = 0.5$ relapses to 33.8%, versus 10.2% for stronger dampening at $\lambda = 0.1$.

Their receptive fields remain close to the baseline (cosine 0.965), and measured threshold decay agrees with e^{-t/τ_θ}. D's synapse gate still relapses to 6.9% on Fashion, whereas its neuron gate removes 99.3% of every targeted neuron's weights and maintains 0% on both datasets. Thus D's neuron gate is the strongest measured option for durable whole-class unlearning; A/C durability was not tested. The result supports the concern that unlearning close to the original weights is vulnerable to relearning [8].

Key finding: relapse In a homeostatic SNN, scaling weights down is reversible: the adaptive threshold revives dampened detectors. Among the audited B/D settings, D's neuron gate gives 0% relapse on both datasets; A/C durability remains unmeasured.

Stretch goal: can we forget individual images? We requested deletion of a random 10% of the 6/9 training images (1,185 MNIST / 1,200 Fashion), while keeping the remaining 90% useful. A cannot target this request because its mask silences entire classes; B/C were evaluated. Successful unlearning should resemble Gold, which never trained on the requested images, while preserving recognition of other 6/9 images. It need not misclassify the forgotten images: Gold can still recognise them through features learned from the retained 90%.

Table 4: Subset stretch goal: common accuracy metrics (%). D_f = the requested 10% of training images; Unseen 6/9 = test images from the same classes; Non-6/9 = test images from the other eight classes. Neither test column measures the retained 90% of 6/9 training images.

	MNIST			Fashion-MNIST		
Model	D_f	Unseen 6/9	Non-6/9	D_f	Unseen 6/9	Non-6/9
Baseline	87.5	88.1	89.5	60.0	59.8	77.6
Gold	87.1[†]	86.2	88.9	64.7[†]	65.6	76.5
B, $\alpha = 10$[‡]	87.4	88.1	89.5	59.8	59.8	77.7
B, $\alpha = 5$	57.4	59.5	90.5	50.5	50.7	79.2
B, $\alpha = 2$	0.1	0.1	92.0	0.3	0.2	80.1
Baseline (C readout)[§]	88.7	88.0	89.2	61.2	61.8	78.0
C1 (subset)[§]	82.6	81.8	90.0	34.2	34.0	78.2
C2 (subset)[§]	78.4	78.6	90.5	33.4	31.4	78.1

[†]Gold's D_f results are from a later diagnostic using saved labels; test results are from its full evaluation. [‡]$\alpha = 25$ reproduces baseline exactly. [§]C uses five passes at $s_u = 0.5$ and retained-only label reassignment (29/55 labels change); compare with its own baseline control. These protocol differences limit direct ranking.

What happened to other 6/9 images? B at $\alpha = 5$ lowers MNIST accuracy on the requested images from 87.5% to 57.4%, but also lowers unseen 6/9 accuracy from 88.1% to 59.5%; at $\alpha = 2$ both fall to about zero. This is damage to the classes, not evidence of targeting only the requested images. C shows the same pattern. In addition to the test results above, its original runs evaluated 200 retained 6/9 training images per dataset: C1/C2 reduce accuracy from 91.0% to 86.0%/79.5% on MNIST, and from 61.0% to 36.0%/32.5% on Fashion. Thus images explicitly meant to be retained also suffer. These retained-training measurements are available for C only and are separate from the common-metric table.

How should closeness to Gold be read? B at $\alpha = 10$ does remain close to Gold on MNIST: D_f accuracy is 87.4% versus 87.1%, and unseen 6/9 accuracy is 88.1% versus 86.2%. However, it also remains almost identical to the baseline; on Fashion its corresponding accuracies are 59.8%/59.8%, versus Gold's 64.7%/65.6%. Closeness in accuracy is compatible with successful unlearning, but does not establish that the requested images' influence was removed. The stronger B settings and five-pass C instead harm other 6/9 images. We therefore found no convincing evidence of selective deletion in these runs; we do not conclude that selective unlearning is impossible or that high forgotten-image accuracy itself means failure.

5 Conclusion

Our starting question was how information is stored in an SNN trained through local learning rules, and whether that organisation makes robust unlearning possible. We expected class-selective neurons to offer more direct targets than the distributed representations commonly associated with

ANNs. The results partly support this idea: silencing or removing the inputs to neurons labelled 6/9 suppresses those classes while largely preserving the rest of the model. However, local weight updates do not make all learned information exclusive to individual neurons. Relabelling reveals that retained-class neurons still recognise 6/9 through shared features, just as Gold does. We did not compare against an ANN directly, so the experiment does not establish that SNNs are inherently easier to unlearn.

The clearest result is that durable forgetting requires more than reducing a detector's activity. B and D both approach Gold's deployed whole-class behaviour, but continued learning can revive B's dampened detectors as their thresholds adapt. D's neuron gate maintains zero removed-class accuracy after 20,000 retained replay images on both datasets, with retained accuracy close to Gold and replay costing at most 3% of retraining. Among the methods audited for durability, it provides the strongest evidence for robust whole-class unlearning. The subset stretch goal exposes a finer-grained challenge: interventions strong enough to affect the requested 10% also damage other images of the same classes, while near-baseline accuracy alone cannot verify deletion.

These findings make targeted unlearning a useful direction for SNNs intended for energy-efficient edge applications, where a deployed model may need to keep learning and respond to deletion requests without full retraining. The local class structure gives us practical intervention points, but shared features and homeostasis determine whether forgetting lasts. Our results support the feasibility of low-cost, durable class removal in this setting; selective image deletion and the actual energy savings on edge hardware remain questions for further work.

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